1. INTRODUCTION

Theory of Structure is one of the serious of courses arranged in the structural engineering module for B.Sc degree program of Construction Technology and Management in the Ethiopian Institute of Architecture, Building Construction and City Development of Addis Ababa University. In this course, Additional methods for the investigation of effects of force on structures will be dealt with the same nature as that of the previous courses: Engineering Mechanics and Strength of Materials. Therefore proper understanding on the topics discussed in the previous courses are found to be basic for smooth continuation and effective understanding of subject maters to be discussed in this course. To this effect, students are advised to do review exercise and refresh their understanding on basic subject matters discussed in Engineering Mechanics and Strength of Materials. In addition introductory discussions on stability and determinacy of structures and loads on structures in more general terms will be discussed in this chapter.

Review of Statics and Strength of Materials:

(To be handled in the form self contained reading and review exercise)

Review Exercise

1. Determine the reaction at supports and axial force in member BD for the pin connected frame loaded as shown in Fig.1.
2. Determine axial force in members BK, JI and EF for the plane truss subjected to loads as shown in Fig.2. Members DI and EJ are flexible cables capable of supporting only tension.
3. Determine the reactions at supports and draw the shear force and bending moment diagrams for the beam loaded as shown in Fig.3.
4. Using direct integration method based on the second order differential equation, find the equation of the elastic curve and specify the slope at B and deflection at C for the beam loaded as shown in Fig.4
5. Using moment area method, determine the deflection at A and the slope at D for the beam loaded as shown in Fig.5
6. Determine the reactions at supports and draw the shear force and bending moment diagrams for the structure under the effect of loads shown in Fig.6

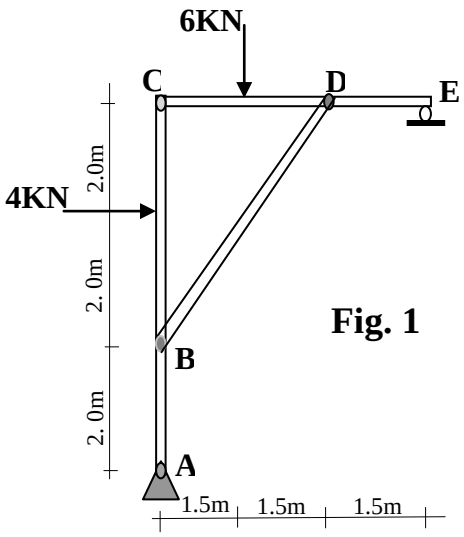


Fig. 1

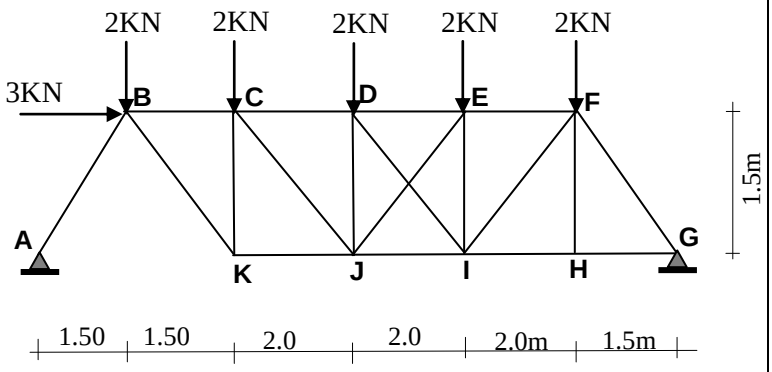


Fig. 2

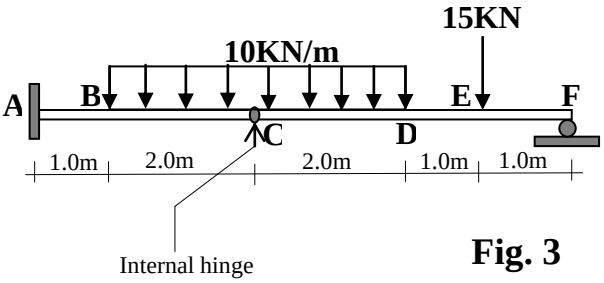


Fig. 3

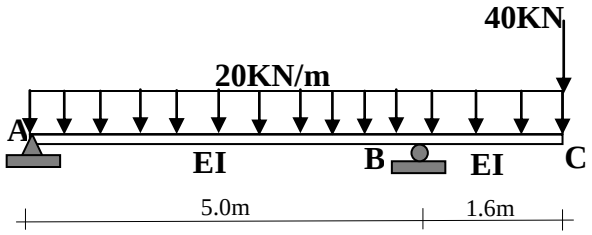


Fig. 4

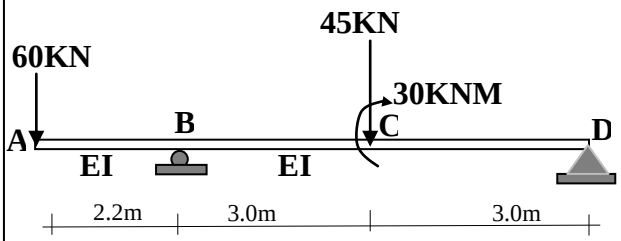


Fig. 5

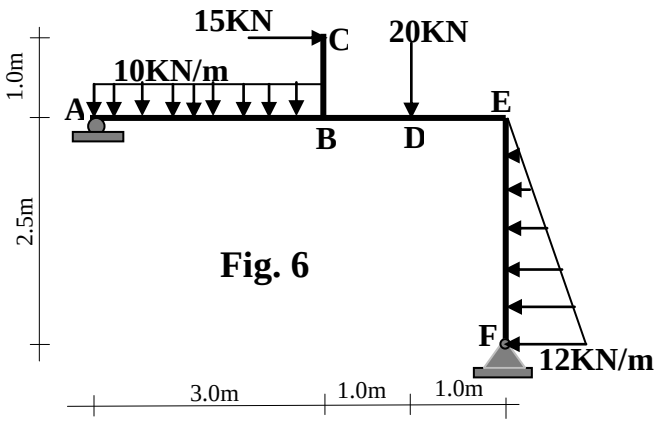


Fig. 6

Stability and Determinacy of Structures:

Stability:

- A structure is said to be stable when it can support any conceivable system of applied loads. Consequently the stability of structures depends on the number and geometrical arrangement of the structural members and supports rather than on the strength of individual members or supports. Therefore the stability of structure in this context is to be assessed only with reference to the number and arrangement of supports and members of structure to be in a stable configuration at all times during the life span of the structure under any conceivable system of loads. As a result the problem of stability associated with excessive deflection and/buckling due to elastic straining are not to be treated in this consideration.

Determinacy:

- A structure is said to be statically determinate if external reactions and internal forces in the members can be determined by using only the independent equilibrium equations. Otherwise it is classified as indeterminate and the degree of indeterminateness equals the number of unknowns minus the number of available equations.

* The stability and determinacy of structures can be discussed from the point of view of
    - External Conditions
    - Internal Conditions
    - Over all Conditions

(i) External Stability and Determinacy:

- It is well known that three reaction components are necessary but not always a sufficient condition for external stability of structures in two dimensions.

  Generally:

  $r_a < r_m$ unstable

  $r_a = r_m$ determinate

$r_a > r_m$ Indeterminate $D = r_a - r_m$

Where :

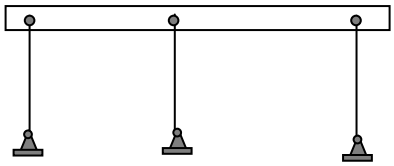
r_a = number of available reaction components

r_m = minimum number of reaction components required for stability.

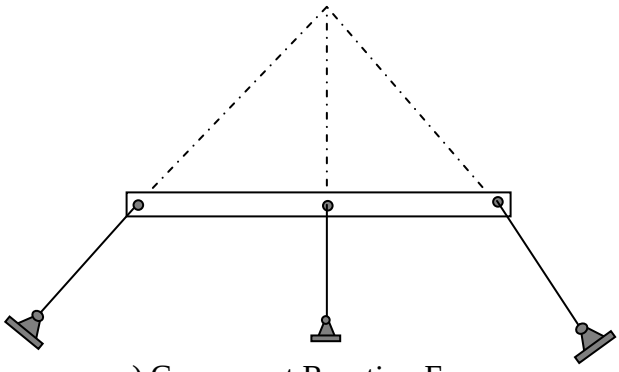
* Note that $r_a \geq r_m$ is not a guarantee for external stability.Arrangement of the reactions or restraints is important.

Example:

The following structure represent typical cases where the systems are geometrically unstable because of improper arrangement of restraints (reaction forces) even though numerically sufficient for stable equilibrium condition in two dimensional force systems.



a) Parallel Reaction Forces



a) Concurrent Reaction Forces

(ii) Internal Stability and Determinacy

Generally a structure is said to be internally indeterminate when it is not possible to determine internal forces in all members of the structure by using only equilibrium equation. If the internal forces in all members of the structure can be determined by using only the static equilibrium equations, it is said to be internally determinate. The internal stability and determinacy of plane truss and continuous rigid frame in two dimensional force systems are discussed subsequently

a) Plane Trusses

It is known that a truss developed by using three bars connected by hinge at their ends in the form of triangle forms stable and internally determinate truss

Assuming externally stable truss supported by minimum number of restraint reaction, r_m; a comparison between the number of unknown forces and number of possible equations can be made as follows:

$2j > m + r_m$	Less members ~ Internally unstable
$2j = m+r_m$	Just sufficient members ~ internally statically determinate
$2j < m+ r_m$	More members ~ Internally statically indeterminate

where :

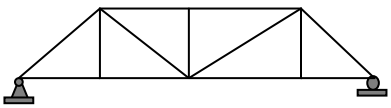
j = number of joints

m=number of members

r_m= minimum number of external reaction components for external stability.

*** Note** that $2j \leq m + r$ is necessary but not sufficient condition for internal stability. It is important to properly inspect the arrangement of members of the truss to form stable triangular panels.

Examples:

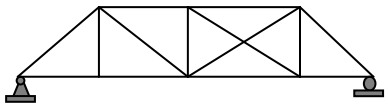


$j = 8, m = 13, r_m = 3$

$2j = 2*8 = 16$

$m+r_m = 13+3 = 16$

⟹ Stable & Determinate

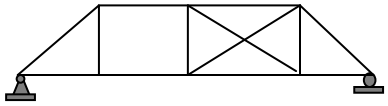


$j = 8, m = 14, r_m = 3$

$2j = 2*8 = 16$

$m+r_m = 14+3 = 17$

⟹ Stable & indeterminate to the 1st degree



$j = 8, m = 13, r_m = 3$

$2j = 2*8 = 16$

$m+r_m = 13+3 = 16$

⟹ Seems determinate but unstable internally

(b) Continuous Rigid Frame Structures

- Continuous frame is a structure composed of rigidly joined members capable of supporting or resisting moments and all force components.
- The following relations can be considered or used to assess the stability and determinacy of framed structures:

$3j+s > 3m+r_m$ the frame is unstable

$3j+s = 3m+r_m$ the frame is determinate

$3j+s < 3m+r_m$ the frame is indeterminate

Where: j = number of rigid joints

m = number of members

r_m= minimum number of support reaction components for external stability

s = number of special condition

* Note that $3j+s \leq 3m+r_m$ is necessary but not sufficient condition for stability. It is important to properly investigate the arrangement of members and occurrence of special conditions so that one or more members of the frame are unstable. It is also equally important to note that the minimum number of reaction r_m for two dimensional frames may not be always three. It may be more depending on the nature of the frame.

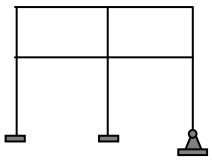
* A classical formula to find the degree of internal indeterminateness is **$D = 3n - I_c$**

Where: D = degree of internal indeterminacy

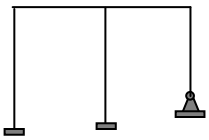
n = No of segments of the area being completely enclosed by frame members

I_c = No of internal conditions

Examples:



j= 9 , m= 10, s= 0, r_m =3
3j+s = 3*9+0 = 27
$3m+r_m$ = 3*10+3 = 33
internally statically indeterminate to the 6th degree and stable.



j = 6, m=5, s = 0, r_m=3

3j+s=3*6+0= 18

$3m+r_m$= 3*5+3 = 18

internally statically determinate & stable.

(iii) Overall /Combined Stability and Determinacy:

When the overall state of stability and determinacy is sought, the combined condition of the structure must be investigated and the following relation holds true:

a) Plane Truss:

$$D = m+r-2j$$

b) Continuous rigid frame Structures:

$$D = (3m+r) – (3j+s)$$

Where:

D = degree of indeterminacy

r = total number of external reactions /constraints

m = number of members

j = number of joints

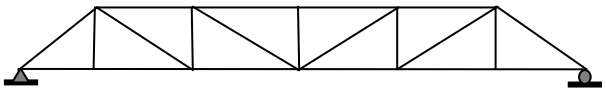
s = number of special condition equations

Exercise

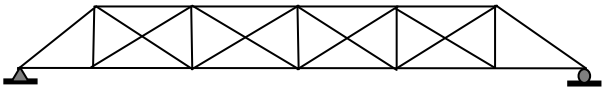
Assess the stability and determinacy of the following structures in terms of external, internal and overall conditions.

1. Plane Trusses

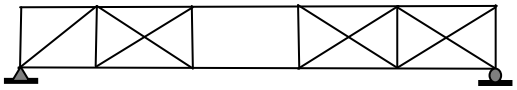
(a)



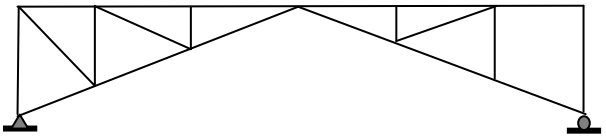
(b)



(c)

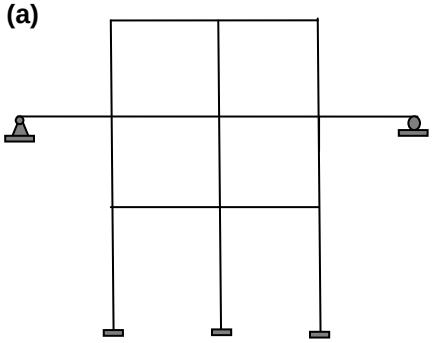


(d)

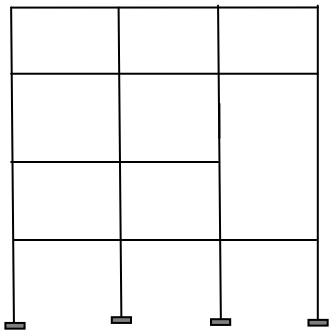


2. Rigid Frames

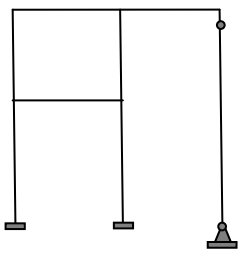
(a)



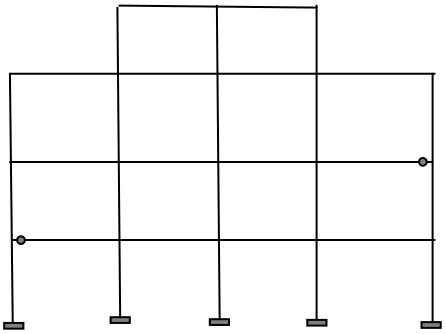
(b)



(c)



(d)



Loads on Structures:

- Loads that possibly act on the structure throughout its intended life time should be estimated in advance and determination of loads represents the first step in the process of analysis and design of structures.

- Loads on structures can be identified in to the following types from the point of view of

i) Direction of Application:
   - Gravity loads
   - Lateral loads

ii) Variation with Time :
   - Dead load
   - Live load
   - Accidental load

iii) Structural Response
   - Static
   - Dynamic:
     - Impact
     - Vibratory
     - Impulsive

Loads to be considered for the design of structures are usually specified in building codes based on function of the structure. For example the Ethiopia Building code standard specifies the following loads as primary loads to be considered for the design of buildings.

Dead Load:

- are those loads to be considered starting at the time of construction and generally remaining more or less with the same magnitude through out the life span of the structure.
- Includes the weight of the structure /components of the structure and that of any permanently attached equipments or fixtures.

Live loads:

- are those loads of temporary nature such as people, distributed furniture, snow etc.
- the design live load depends on the building's /structure's intended use and generally considered as equivalent uniformly distribute static load over floor area.

Wind loads:

- wind load on a structure result from a change of velocity of the air due to obstruction of the structure , friction b/n the air and the structure and possible aerodynamic forces.

Earth Quake loads:

Earth quake loads are forces resulting from the acceleration of the supporting ground. Movement of ground during earth quake can be in any arbitrary direction but the most damaging effect on building structure is the movement in the horizontal direction. Lateral force during earth quake basically depends on the self weight/mass of the building and extent of ground acceleration and are normally regarded as inertia forces. The governing parameters; type and extent of ground motion based on tectonic events and geological history of the locality are normally given in building codes. Using such parameters and mass of the building, pertinent lateral forces should be computed to act in the lateral direction for analysis and design of the building. Earth quake forces can be considered in any conceivable direction for the design of the structure. But at least two orthogonal directions must be considered based on the orientation of the structural system.